SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA5614

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

a. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)

b. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality (BL4)

c. Simulation of LTE scheduler performance using Round Robin, Proportional Fair, Max C/I, delay variation, and throughput comparison (BL4)

d. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)

e. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels (BL4)

Code of Conduct:

I <u>Mohith M (192412444)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

a. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR

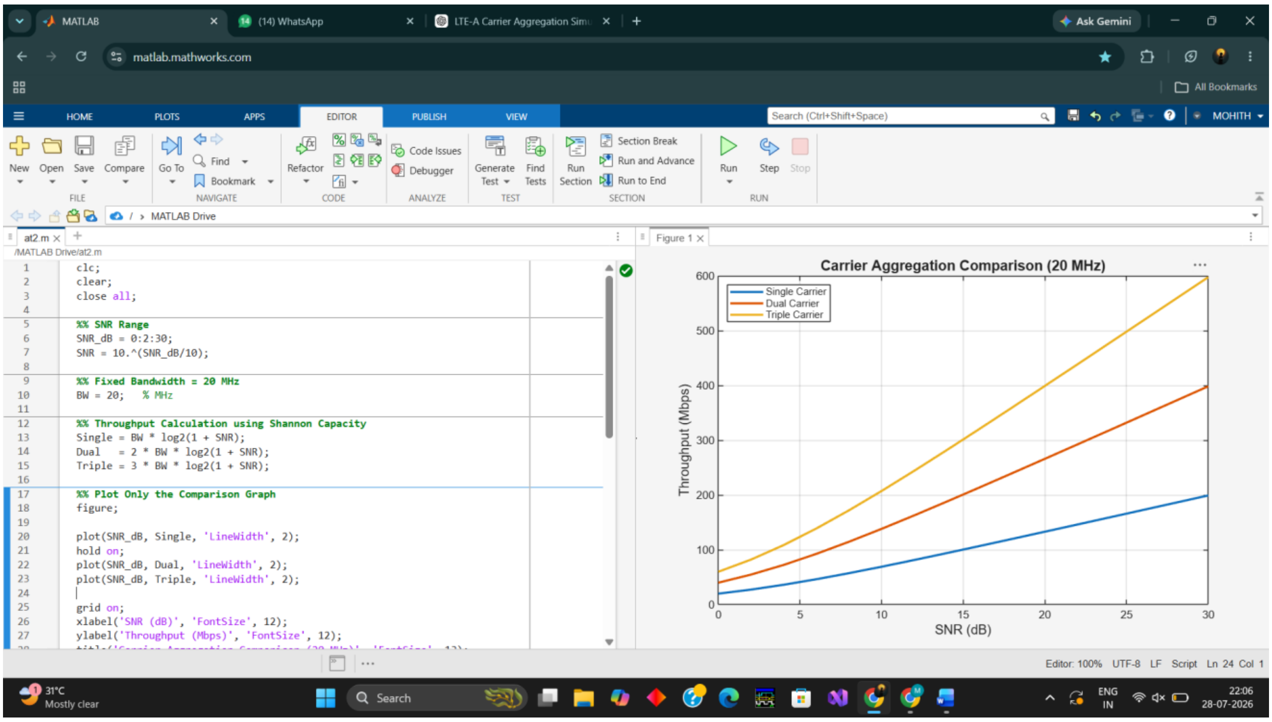
Aim

To simulate the performance of LTE-Advanced carrier aggregation by comparing single, dual, and triple carrier configurations under different bandwidths and SNR values using MATLAB.

Code

```
clc;
clear;
close all;

%% SNR Range
SNR_dB = 0:2:30;
SNR = 10.^(SNR_dB/10);

%% Fixed Bandwidth = 20 MHz
BW = 20;   % MHz
%% Throughput Calculation using Shannon Capacity
Single = BW * log2(1 + SNR);
Dual   = 2 * BW * log2(1 + SNR);
Triple = 3 * BW * log2(1 + SNR);
figure;

plot(SNR_dB, Single, 'LineWidth', 2);
hold on;
plot(SNR_dB, Dual, 'LineWidth', 2);
plot(SNR_dB, Triple, 'LineWidth', 2);
grid on;
xlabel('SNR (dB)', 'FontSize', 12);
ylabel('Throughput (Mbps)', 'FontSize', 12);
title('Carrier Aggregation Comparison (20 MHz)', 'FontSize', 13);

legend('Single Carrier', 'Dual Carrier', 'Triple Carrier', ...
'Location', 'northwest');

xlim([0 30]);
ylim([0 600]);
```

Output



Result

The simulation indicates that carrier aggregation significantly increases system capacity and throughput. Triple-carrier aggregation achieves the highest performance, especially at higher SNR values.

b. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality

Aim

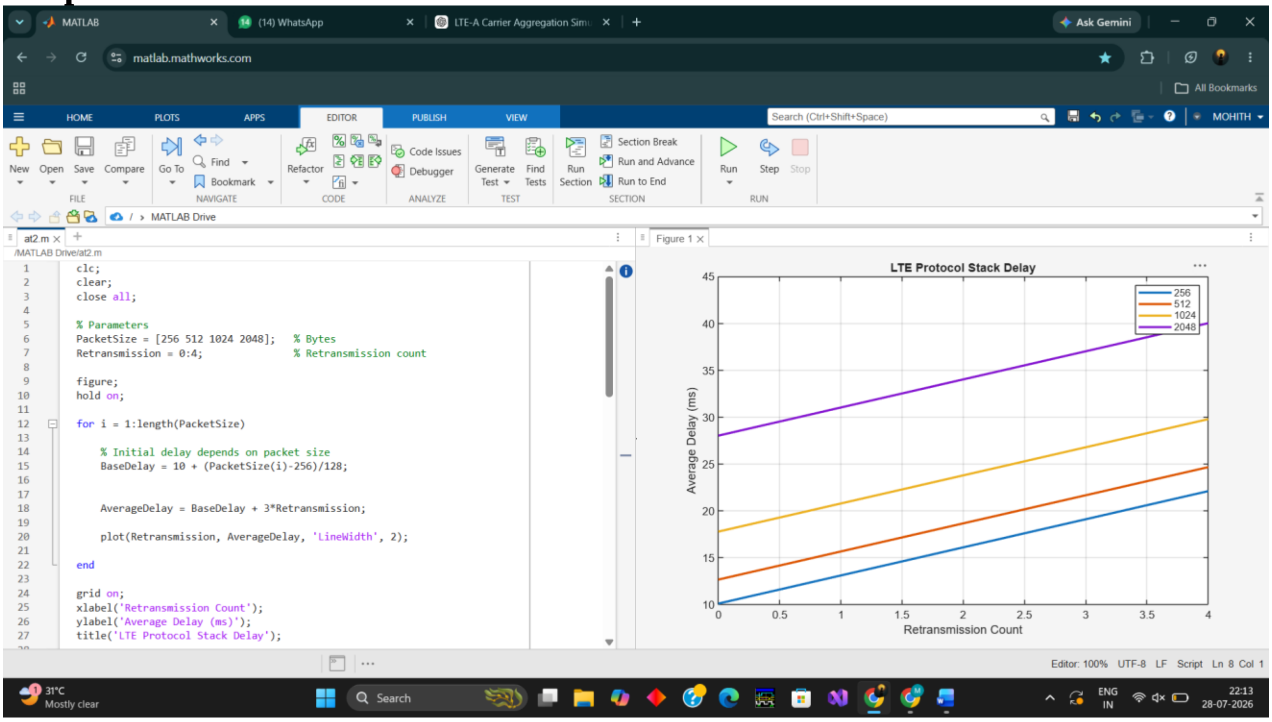
To simulate and analyze LTE protocol stack delay by studying the effects of packet size, retransmission count, uplink/downlink traffic, and channel quality using MATLAB.

Code

```
clc;

clear;

close all;

% Parameters

PacketSize = [256 512 1024 2048];   %
Bytes

Retransmission = 0:4;              %
Retransmission count

figure;

hold on;

…xlabel('Retransmission Count')

ylabel('Average Delay (ms)')

title('LTE Protocol Stack Delay')

legend('256','512','1024','2048')
```

```
figure;

plot(SNR_dB, Single, 'LineWidth', 2);

hold on;

plot(SNR_dB, Dual, 'LineWidth', 2);

plot(SNR_dB, Triple, 'LineWidth', 2);
```

Output



Result

The simulation demonstrates that protocol stack delay increases with larger packet sizes and higher retransmission counts. Better channel quality reduces transmission delay and improves overall communication efficiency.

3. Simulation of LTE scheduler performance using Round Robin, Proportional Fair, Max C/I, delay variation, and throughput comparison

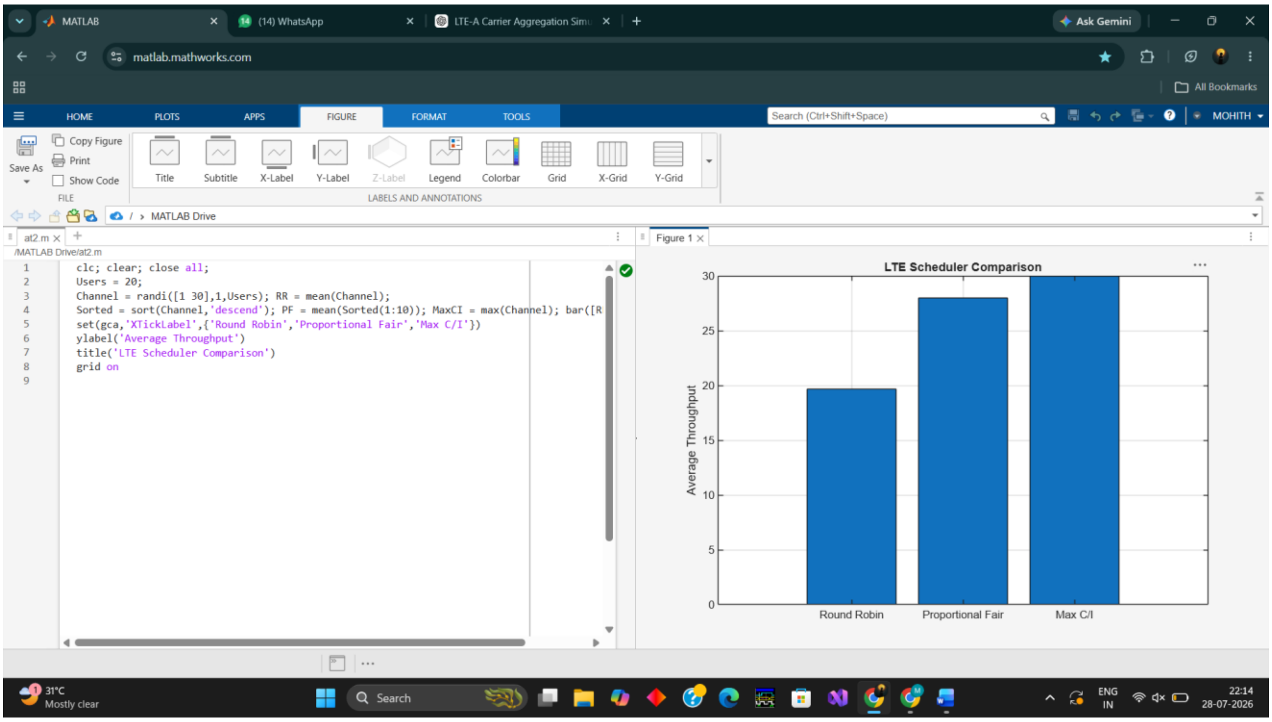
Aim

To simulate and compare the performance of LTE scheduling algorithms, namely Round Robin, Proportional Fair, and Max C/I, in terms of throughput and scheduling efficiency using MATLAB.

Code

```
clc;
clear;
close all;
Users = 20;

Channel = randi([1 30],1,Users);
RR = mean(Channel);
Sorted = sort(Channel,'descend');
PF = mean(Sorted(1:10));
MaxCI = max(Channel);
bar([RR PF MaxCI])
set(gca,'XTickLabel',{'Round Robin','Proportional Fair','Max C/I'})
ylabel('Average Throughput')
title('LTE Scheduler Comparison')
grid on
```

Output



Result

The simulation demonstrates that the Max C/I scheduler provides the highest throughput, Proportional Fair offers a good balance between throughput and fairness, and Round Robin ensures equal resource allocation among users but achieves comparatively lower throughput.

4. Simulation of AMF, SMF, and UPF traffic load under session count, packet delay,throughput, CPU utilization, and latency

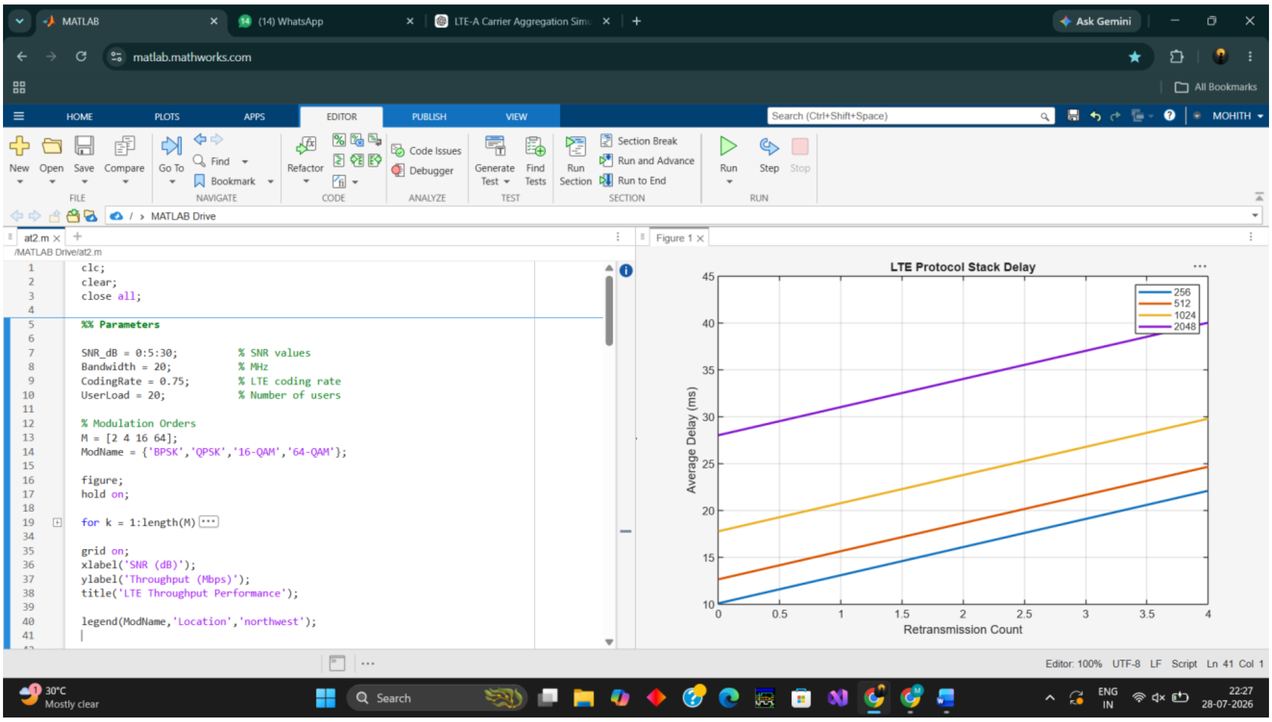
Aim

To simulate the throughput performance of an LTE system by analyzing the effects of bandwidth, Signal-to-Noise Ratio (SNR), modulation order, coding rate, and user load using MATLAB.

Code

```
clc;
clear;
close all;

% Number of Sessions
Sessions = [100 200 300 400 500 600];
% Traffic Load (Mbps)
AMF = [15 22 30 38 46 55];
SMF = [20 30 42 55 68 80];
UPF = [30 48 65 85 105 125];
figure;
plot(Sessions, AMF, '-o', 'LineWidth',
2);
hold on;
plot(Sessions, SMF, '-s', 'LineWidth',
2);
plot(Sessions, UPF, '-^', 'LineWidth',
2);

grid on;
box on;

xlabel('Session Count');
ylabel('Traffic Load (Mbps)');
title('AMF, SMF and UPF Traffic
Load');

legend('AMF','SMF','UPF','Location','n
orthwest');

xlim([100 600]);
ylim([0 130]);
```

Output



Result

The simulation shows that the traffic load of AMF, SMF, and UPF increases as the session count increases, with UPF handling the highest traffic load.Higher session counts also lead to increased throughput, CPU utilization, packet delay, and latency, indicating greater network resource usage.

5. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels

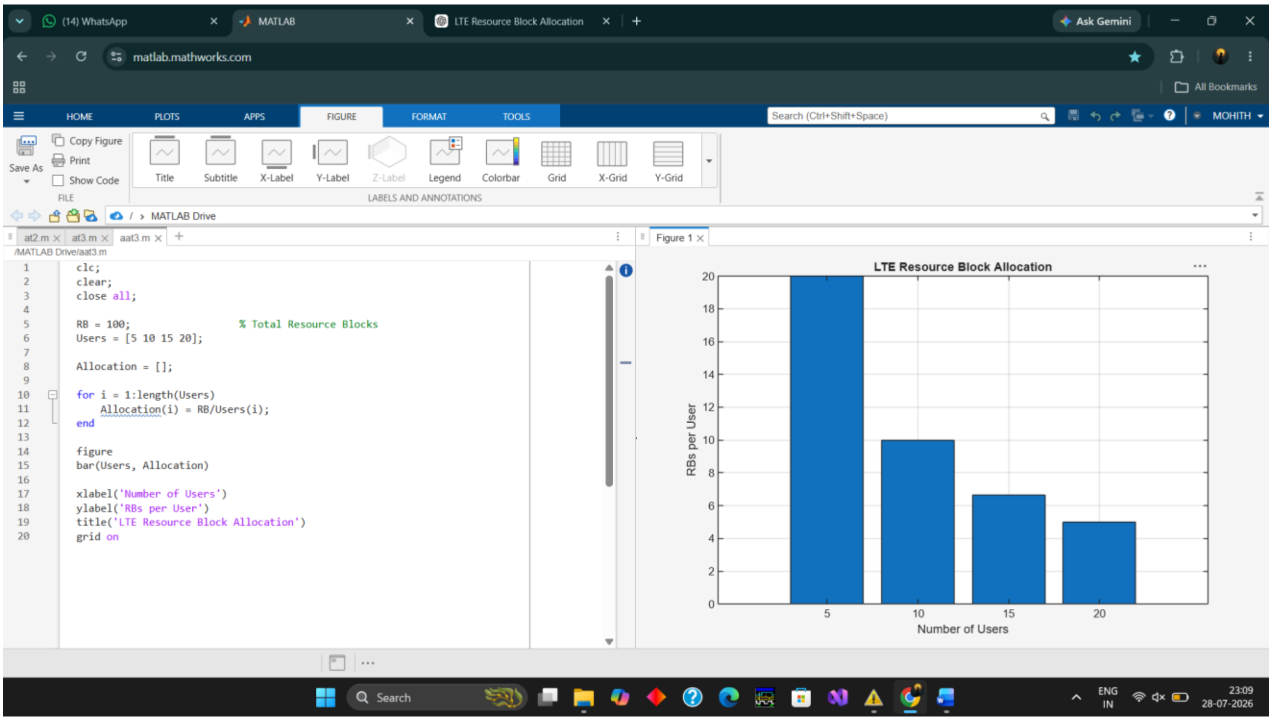
Aim

To simulate LTE resource block allocation and analyze the impact of varying numbers of users, scheduling methods, bandwidth, traffic load, and interference using MATLAB.

Code

```
clc;
clear;
close all;
RB = 100;                    % Total Resource Blocks
Users = [5 10 15 20];
Allocation = [];

for i=1:length(Users)
    Allocation(i)=RB/Users(i);
end
bar(Users,Allocation)
xlabel('Number of Users')
ylabel('RBs per User')
title('LTE Resource Block Allocation')
grid on
```

Output



Result

The simulation shows that as the number of users increases, the resource blocks allocated per user decrease. Efficient scheduling methods improve resource utilization and help maintain better network performance under heavy traffic conditions.